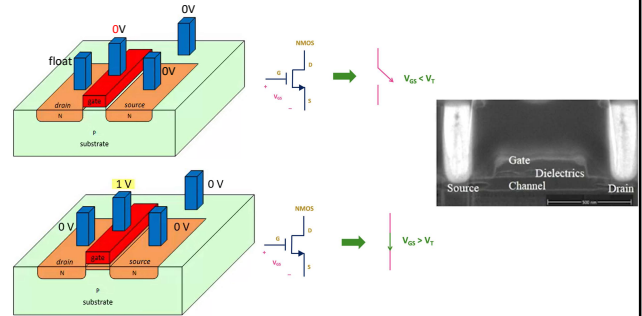
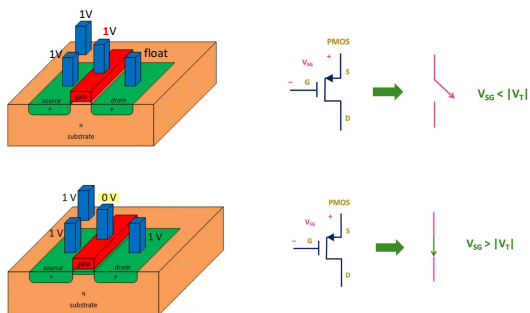


Functionality and Manufacturing of ICs

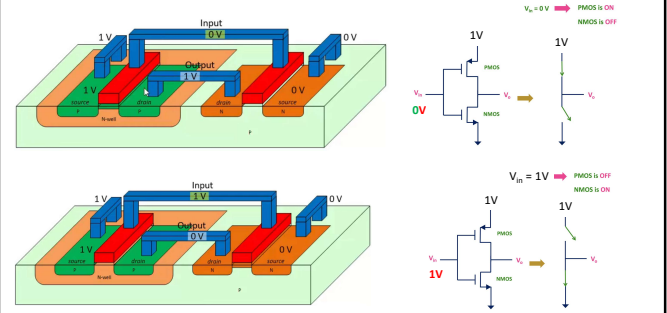
NMOS Transistor



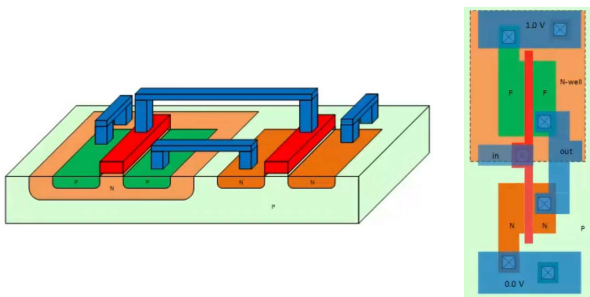
PMOS Transistor



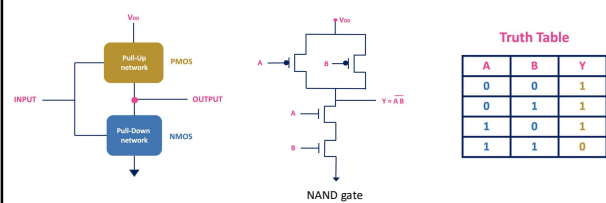
CMOS Inverter



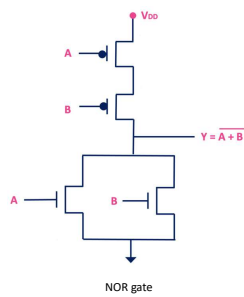
CMOS inverter: layout



CMOS gates



CMOS gates

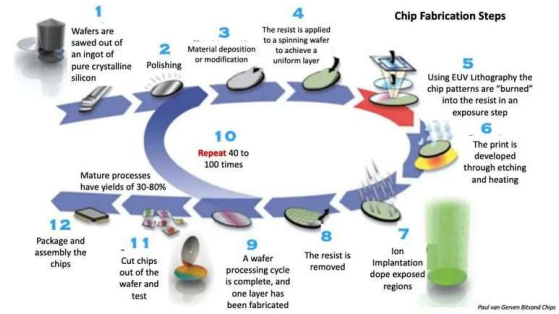


Truth Table

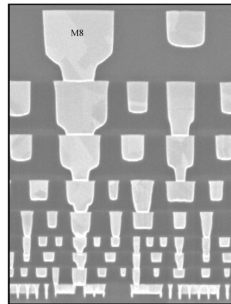
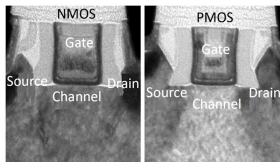
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NOR gate

IC Fabrication



Transistor and Interconnects

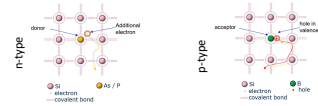


VLSI fabrication technology

Silicon ingot: very high purity, single-crystal silicon.

Wafer: an ingot is sawed to produce a circular wafer (400-600 um thick). The surface of the wafer is polished using chemical and mechanical polishing (CMP) techniques.

A specific amount of impurities can be added to the pure silicon in a process known as **doping**. Depending on the types of impurity, either holes (in p-type silicon) or electrons (in n-type silicon) can be responsible for electrical conduction.



Silicon Ingot

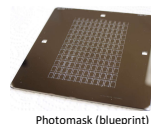
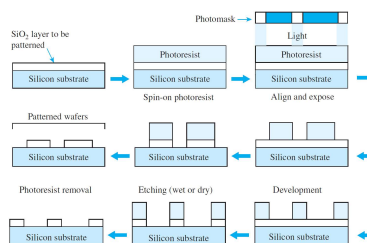


Wafer

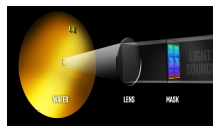
Oxidation: silicon reacts with oxygen to form silicon dioxide. Oxidation is carried out at high temperatures (e.g., 1000–1200°C) inside ultraclean furnaces.

VLSI fabrication technology

The surface patterns of the various integrated-circuit components can be defined repeatedly using **photolithography**



Photomask (blueprint)



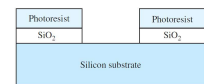
<https://www.youtube.com/watch?v=u3w50UebnSE>

VLSI fabrication technology

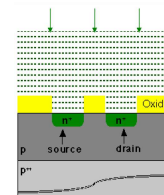
Etching (wet or dry) is used to remove the oxide where the photoresist pattern is absent. The photoresist is then stripped completely off the wafer, leaving the oxide pattern on the wafer.

Wet etching: hydrofluoric (HF) acid to etch SiO_2 , potassium hydroxide (KOH) for silicon, phosphoric acid for aluminium

Dry etching: use a corrosive gas (or ions)

**Ion Implantation:**

- introduces impurities into the semiconductor crystal.
- An ion implanter produces ions of the desired dopant, accelerates them by an electric field, and allows them to strike the semiconductor surface. The ions become embedded in the crystal lattice.
- The depth of penetration is related to the energy of the ion beam, which can be controlled by the accelerating-field voltage.



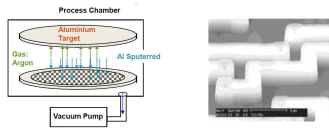
VLSI fabrication technology

Chemical vapor deposition (CVD) is a process in which gases or vapors are chemically reacted, resulting in the formation of solids on a substrate.

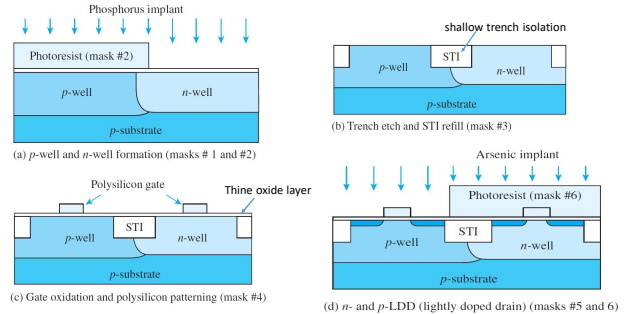
CVD can deposit SiO_2 , Si_3N_4 , and polysilicon on a silicon substrate.

Metallization:

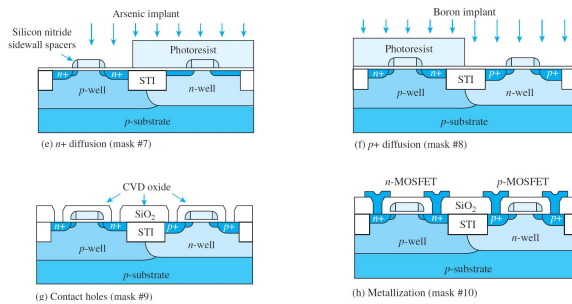
- Interconnect the various components to form the desired integrated circuit.
- Deposits metal over the entire surface of the silicon.
- The required interconnection pattern is then selectively etched.
- The metal layer is normally deposited via a sputtering process.



VLSI processes



VLSI processes



VLSI layout

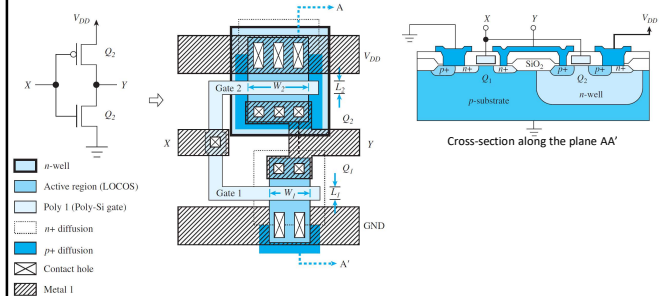


Photo mask

